

High-gain and Low-noise Integrated Fiber Amplifier with Seamless 102 nm Bandwidth Across the Extended C+L Band

Xin Huang⁽¹⁾, Junjie Qi⁽¹⁾, Shiqi Zhou⁽¹⁾, Zhaolong Liao⁽¹⁾, Yuanpeng Ding^(1,2), Chan Li⁽¹⁾, Xiaoying Nie⁽¹⁾, Yuan Gao⁽¹⁾, Hui Zhang⁽¹⁾, Zhixiong He⁽¹⁾, Lei Shen^(1,*), Lei Zhang⁽¹⁾, Jie Luo⁽¹⁾

⁽¹⁾ State Key Laboratory of Optical Fiber and Cable Manufacture Technology, Yangtze Optical Fiber and Cable Joint Stock Limited Company (YOFC), Wuhan 430073, China, *shenlei@yofc.com

⁽²⁾ Wuhan National Laboratory for Optoelectronics and School of Optical and Electronic Information, Huazhong University of Science and Technology, Wuhan 430074, Hubei, China

Abstract We report a hybrid EBDF/BDF fiber amplifier that achieves integrated amplification with a seamless 102 nm spectral bandwidth in the extended C+L band for the first time. The amplifier exhibits a gain of 26.7 dB and a noise figure of 4.6 dB. ©2026 The Author(s)

Introduction

The rapid development of emerging network applications, including cloud computing, artificial intelligence, and virtual reality, has put unprecedented demands on the capacity of communication systems [1]. Consequently, expanding the communication bandwidth from the C band to the C+L band has become imperative. Constrained by existing erbium-doped fiber (EDF) technology, it remains challenging to achieve simultaneous amplification of C+L band signals within a single erbium-doped fiber amplifier (EDFA). Thus, traditional full C+L band amplification typically employs independent C and L band EDFAs. However, the inherent bandwidth constraints of individual EDFAs inevitably introduce spectral gaps between the C and L bands, resulting in substantial wastage of valuable spectral resources. And the introduction of C/L band wavelength division couplers also increases additional system losses, leading to system complexity and performance degradation, high power consumption, and high cost [2].

A single integrated C+L band fiber amplifier has been proposed to overcome the shortcomings of current traditional transmission scheme. Due to its unique ultra-wideband luminescence characteristics, bismuth ions can cover an operating wavelength of 1150-1800 nm [3-5], making erbium-bismuth co-doping fiber (EBDF) technology a core strategy for realizing integrated amplification in the C+L band. Researchers have built a single-stage C+L band erbium-bismuth co-doped fiber amplifier (EBDFA), achieving a 20 dB gain bandwidth of 67 nm [6]. Another study reported a double-pass C+L band amplifier constructed by EBDF and long-period fiber grating, which can extend the 20 dB gain bandwidth to 85 nm [7]. Besides, a C+L+U band EBDFA has achieved a 15 dB small-signal gain bandwidth covering 260 nm (1515-1775 nm) [8]. Based on a bidirectional pump structure, an EBDFA achieved a gain of over 11.8 dB and a noise figure (NF) of 9.3 dB in the range of 1530-1670 nm [9]. Above

results show that integrated C+L band amplifiers still face the technical bottlenecks of high NF and difficulty in fully covering the extended C and L bands (1524-1627 nm) with a 20 dB gain bandwidth. Therefore, designing and developing an integrated fiber amplifier with high gain, low NF, and bandwidth that can cover the full extended C+L band is of great value and practical application significance.

In this paper, we successfully fabricate the EBDF and bismuth-doped fiber (BDF) using the modified chemical vapor deposition (MCVD) platform. Furthermore, we innovatively develop an integrated extended C+L band fiber amplifier via a cascaded amplification scheme. The excellent amplification performance of the integrated amplifier is achieved by experimentally optimizing the cascaded amplification optical path parameters. With the injection of 125-channel signals across the full seamless extended C+L band, the amplifier achieves a gain of over 23.6 dB from 1524.1 nm to 1626.4 nm, an average gain of 26.7 dB, an average NF of 4.6 dB, and a saturated output power of 23.2 dBm. This study reveals the significant potential of integrated C+L band fiber amplifiers for the application in C+L band systems.

Fiber characteristics and experimental setup

In this study, the EBDF and BDF are fabricated in-house using the MCVD technique combined with solution doping. The core and cladding diameters of EBDF are 5.7 μm and 124.5 μm . While the BDF exhibits a core diameter of 4.6 μm and a cladding diameter of 124.6 μm , respectively. The peak absorption coefficient of EBDF at 1530 nm is 7.5 dB/m. The BDF contains silicon, bismuth, and germanium (Ge) elements, and its gain performance in the L+U band is mainly attributed to the optimization and enhancement of the concentration of bismuth active centers

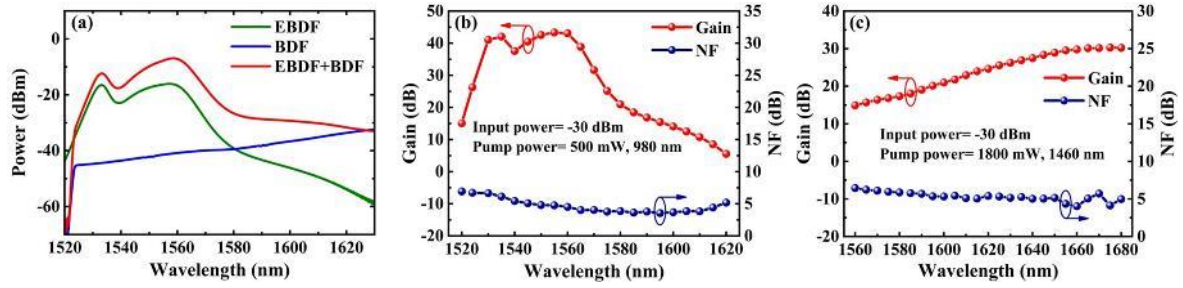


Fig. 1: (a) ASE spectra of EBDF, BDF and EBDF-BDF; (b) The gain and NF curves of EBDF; (c) The gain and NF curves of BDF.

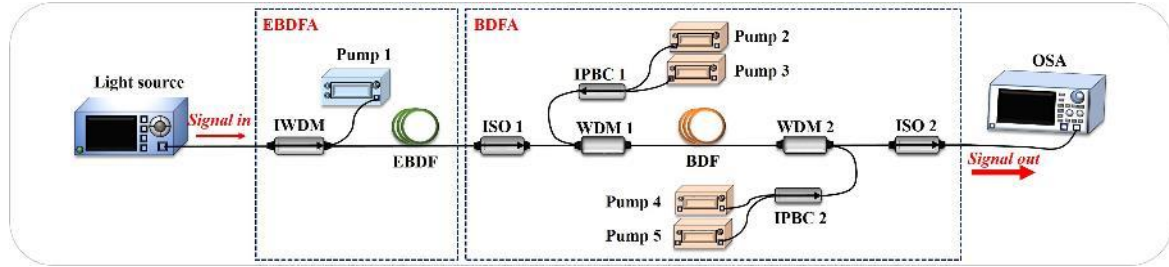


Fig. 2: Experimental architecture of the extended C+L band integrated amplifier.

(BAC)-Ge, the GeO₂ content is estimated as 30~40 mol%. To investigate the amplification characteristics of EBDF and BDF, we utilize a commercial optical spectrum analyzer (OSA) to measure the amplified spontaneous emission (ASE), gain and NF performance. The ASE spectra in Fig. 1(a) indicate that the main operating wavelengths of the EBDF are in the C-band and the short-wavelength region of the L-band. And the ASE spectrum of BDF primarily covers the L+U band, which is crucial for compensating the insufficient gain at longer wavelengths in the integrated C+L band amplifier. It can be clearly seen that the ASE spectrum characteristics of the EBDF-BDF cascaded amplification is the superposition effect of the ASE spectra of the two fibers alone. The corresponding gain and NF curves of the EBDF and BDF at a signal power of -30 dBm are presented in Fig. 1(b) and Fig. 1(c). The EBDF results indicate that within the wavelength range of 1520-1595 nm, the gain exceeds 15 dB while the NF remains below 6.9 dB. Across the wavelength range exceeding 1565-1680 nm, the BDF provides a gain exceeding 15 dB with an NF less than 6.2 dB. Specifically, a gain of 25.5 dB and an NF of 5.3 dB are achieved at 1625 nm in the long-wavelength region of the L-band.

The integrated optical amplifier structure, illustrated in Fig. 2, enables seamless extended C+L band amplification by cascading a BDF amplification stage after the EBDF amplification stage. In this setup, the first-stage EBDF employs a 980 nm pump source with 600 mW output power. The amplified C+L band signals from the EBDF enter the bismuth-doped fiber amplifier (BDFA) stage for secondary amplification. In the BDFA stage, a bidirectional pumping scheme is adopted to improve amplification performance.

Two polarization-maintaining (PM) 1460 nm pump lasers with 500 mW output power are used in both the forward and backward directions. Specifically, Pump 2 and Pump 3 are combined via an integrated isolator and polarization beam combiner (IPBC 1) as forward pumps, while Pump 4 and Pump 5 are combined through another IPBC 2 as backward pumps. Finally, the total output is connected to a commercial OSA for characterizing and monitoring the amplified signals.

Results and discussion

In our experiments, the amplification characteristics of the extended C+L band integrated amplifier are investigated at a single-wavelength input power of -30 dBm. The lengths of the EBDF and BDF are set to 16 m and 500 m, respectively. By flexibly adjusting the pump power combination parameters of the EBDF-BDF cascaded amplification configuration, high gain performance across the full C+L band is achieved. Specifically, Figure 3(a) shows the amplified output spectra of different single-wavelength signals, while Fig. 3(b) presents the gain and NF curves. The experimental results indicate that the variation trend of output spectrum with wavelength aligns well with the gain profile, confirming the accuracy of the test results. The integrated fiber amplifier achieves an average gain of 34.9 dB and an average NF of 4.9 dB across the full extended C+L band (1524.1-1626.8 nm), with gains all exceeding 22.5 dB. Furthermore, the signal gains and NFs as functions of input signal powers are measured at three typical wavelengths (1524.1 nm, 1558.7 nm, and 1626.8 nm), as shown in Fig. 3(c). The results show that as the input power increases, the signal gain decreases while the NF

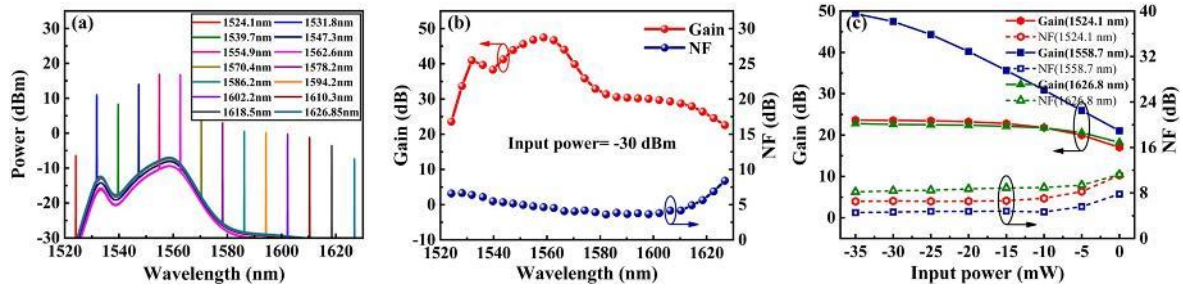


Fig. 3: Single-wavelength amplification characteristics. (a) Output spectra; (b) Gain and NF curves; (c) Gain and NF curves for three typical wavelengths at different input powers.

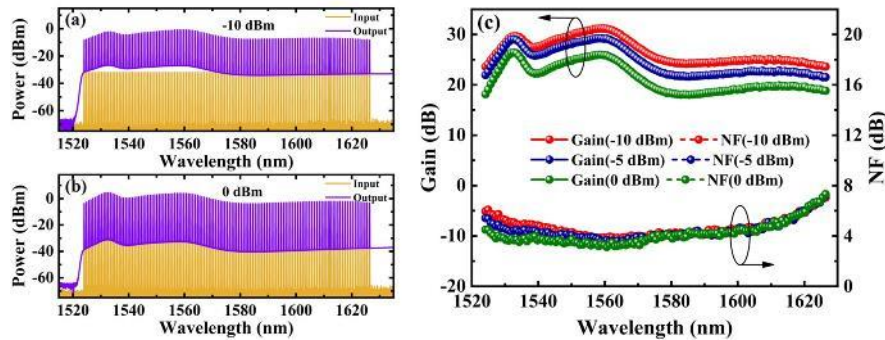


Fig. 4: Multi-wavelength amplification characteristics. (a) Input and output spectra at a signal input power of -10 dBm; (b) Input and output spectra at a signal input power of 0 dBm; (c) Gain and NF curves at different input powers.

risers. Specifically, when the input power is adjusted from -35 dBm to 0 dBm at 1558.7 nm, the gain decreases from 49 dB to 21 dB, and the NF increases from 4.5 dB to 7.8 dB. At the two edge wavelengths of the extended C+L band, the gain decreases slowly as the input power increases. With a 0 dBm signal injection, the gains at 1524.1 nm and 1626.8 nm can reach 17 dB and 18.2 dB, respectively.

In addition, we employ a 125-channel multi-wavelength light source system (ranging from 1524.1 nm to 1626.4 nm with a 100 GHz spacing) to test the multi-wavelength amplification performance of the extended C+L band integrated fiber amplifier. By optimizing the combination of fiber lengths and pump power ratio, the gains of short and long wavelengths in the C+L band are balanced, while simultaneously achieving high gain and low NF performance for full extended C+L band signals. In the experiment, the amplification performance of a 125-channel wavelength-division multiplexing (WDM) signal is evaluated under different input powers. Figure 4(a) and Figure 4(b) present the input and amplified output spectra when the total input power is set to -10 dBm and 0 dBm respectively. Figure 4(c) shows the corresponding gain and NF curves for total input powers of -10 dBm, -5 dBm, and 0 dBm, respectively. The final experimental results demonstrate that with a 125-channel WDM signal input power of -10 dBm, the integrated fiber amplifier achieves a gain exceeding 23.6 dB across the entire extended C+L band (1524.1 nm-1626.4

nm), with an average gain of 26.7 dB. The NF remains below 7 dB over the full band, with an average value of 4.6 dB. At an input power of -5 dBm, this amplifier can achieve an average gain of 24.6 dB and an average NF of 4.4 dB. When the total input signal power is 0 dBm, the average gain is more than 21.4 dB and the average NF is 4.1 dB. Besides, the saturated output power of the amplifier for the full-channel signal exceeds 23.2 dBm.

Conclusion

In summary, by utilizing self-developed EBDF and BDF, we have successfully designed and constructed an extended C+L band integrated fiber amplifier. Distinct from conventional discrete EDFA schemes, this work notably achieves seamless integrated amplification with a 102 nm spectral bandwidth across the full extended C+L band. Experimental results demonstrate that within the wavelength range of 1524.1 nm to 1626.4 nm, the amplifier delivers a multi-wavelength signal gain exceeding 23.6 dB, an average NF of 4.6 dB, and a saturated output power of over 23.2 dBm for full-channel signals. These results not only provide valuable guidance for the subsequent design of amplifier gain flattening and the optimization of multi-stage amplification, but also further highlight the significant application potential of the C+L band integrated fiber amplifier in practical optical communication systems.

Acknowledgements

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